

GSEB CLASS 10 SOCIAL SCIENCE

Chapter 5: Indian Science and Technology - Section B

SECTION B

Q1. Write the contribution of India in the field of Astrology.

- **Development:** India made important contributions to the study of **astrology and astronomy** in ancient times.
- **Varahmihir:** Varahmihir divided astrology into **Tantra, Hora and Samhita**.
- **Important Work:** He wrote **Brihat Samhita**, which contains information related to planets, human life and auspicious occasions.
- **Astronomical Contribution:** Indian scholars such as **Aryabhata** and **Brahmagupta** also made important contributions to astronomy.

OR

Describe Varahmihir's contribution into Astrology.

- **Great Scholar:** Varahmihir was a great **astronomer and astrologer** of ancient India.
- **Three Divisions:** He divided astrology into **Tantra, Hora and Samhita**.
- **Brihat Samhita:** He wrote the famous **Brihat Samhita**, an important work on astrology and related subjects.
- **Contents:** It discusses the effects of celestial bodies, human characteristics and auspicious times for **marriage, digging wells and ponds, gardening and sowing crops**.

Q2. Write the contribution of Nagarjuna in the field of chemistry.

- **Acharya Nagarjuna:** He was a Buddhist scholar of **Nalanda** and is regarded as an important acharya of Indian chemistry.
- **Books:** He wrote works such as **Rasaratnakar** and **Arogyamanjari**.
- **Medicines:** He advised the use of **chemical medicines** along with herbal medicines.
- **Mercury:** He is credited in the textbook tradition with introducing the medicinal use of **mercury ash (bhasma)**.

Q3. State the importance of science and technology.

- **Rich Heritage:** Ancient India had a rich heritage of **science and technology**.
- **Various Fields:** Important contributions were made in **chemistry, metallurgy, mathematics, medicine, surgery, astrology and astronomy**.
- **Practical Use:** Scientific knowledge was used in **medicines, metalwork, architecture** and other practical activities.
- **Historical Importance:** These achievements show the advanced scientific knowledge and skills of **ancient Indians**.

Q4. What are the distinct features of the Indian culture?

- **Unity in Diversity:** Indian culture shows **unity in diversity** despite differences among communities.
- **Tolerance:** It respects different **religions, beliefs, traditions and ways of life**.
- **Harmony:** It combines **religion and science, traditional ideals and practical knowledge**.
- **Cultural Synthesis:** Different communities have lived together and contributed to India's rich and composite culture.

Q5. State the distinct feature of Nalanda University with reference to chemistry.

- **Chemistry Centre:** Nalanda University had facilities for the **study and research of chemistry**.
- **Separate Laboratory:** It had a separate **chemistry laboratory** for chemical studies and research.
- **Furnaces:** Special **furnaces** were also available for chemical experiments.
- **Advanced Knowledge:** This shows the advanced level of **chemical knowledge and experimental practices** in ancient India.